

# Cyvero: The Cyber Security School

## Academic Syllabus: Cybersecurity with AI Mastery Program

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### Module 0.1: Foundations of Networking

#### Learning Objectives:

This module provides a comprehensive introduction to the fundamental principles of networking, with a focus on the theoretical and practical aspects of data communication.

#### Topics Covered:

- **OSI & TCP/IP Models**
  - The seven layers of the OSI model
  - The four-layer TCP/IP model
  - Concepts of encapsulation and decapsulation
  - Comparative analysis of OSI and TCP/IP models
- **IP Addressing**
  - Fundamentals of IPv4 and IPv6
  - Differentiation between public and private IP addresses
  - Classification of IP addresses (Classes A, B, C, D, E)
  - Identification of Network and Broadcast IDs
  - Static vs. Dynamic IP addressing
- **Subnetting**
  - Basics of subnet masks
  - Subnetting techniques for Class A, B, and C networks
  - CIDR (Classless Inter-Domain Routing) notation
  - Practical subnetting problems

- **Core Protocols**

- DHCP: Server and client operations, DORA process
- DNS: Hierarchy, forward and reverse lookup, record types (A, MX, PTR)
- ARP: Request and reply processes
- ICMP: Message types, Ping and Traceroute utilities

- **Routing**

- Static routing and default routes
- Dynamic routing protocols (RIP)
- Routing tables, metrics, and administrative distance

- **Access Control Lists (ACL)**

- Standard and Extended ACLs
- Rules for ACL placement (inbound and outbound)
- Real-world scenarios and troubleshooting techniques

- **Cryptography**

- Symmetric vs. Asymmetric encryption
- Hashing algorithms (MD5, SHA)
- Digital signatures
- Public Key Infrastructure (PKI) and certificates

- **VPN Technologies**

- Site-to-Site VPN
- Remote Access VPN
- Basics of IPsec
- Tunneling protocols (GRE, L2TP, SSL VPN)

- **Wireless Security**

- Wi-Fi 802.11 standards
  - Comparison of WPA2 and WPA3
  - Common wireless attacks and mitigation strategies
  - Techniques for securing wireless networks
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## **Module 0.2: Pre-Security Fundamentals**

### **Learning Objectives:**

This module establishes the foundational knowledge required for understanding security concepts, with an emphasis on operating systems, networking, and security principles.

### **Topics Covered:**

- **Linux Basics**

- System architecture
- File system hierarchy (FHS)
- Basic terminal commands

- Setup of VirtualBox and VMware
- **Users & Permissions**
  - File permissions (read, write, execute)
  - Usage of chmod and chown commands
  - User creation and management
  - Process management tools (htop, ps, top)
- **Linux Networking**
  - IP configuration (ifconfig, ip)
  - Network utilities (netstat, ss, ping, curl, grep, tail)
  - Monitoring authentication logs and SSH attempts
- **Windows Fundamentals**
  - Windows system architecture
  - NTFS file system
  - Services and Registry
  - Task Manager, regedit, and msconfig utilities
- **Windows Security**
  - User Account Control (UAC)
  - Windows Defender and Firewall
  - Password policies
  - Security tools: secpol.msc, gpedit.msc
- **Windows Logs**
  - Event Viewer
  - Critical Event IDs: 4624 (login), 4625 (failed login), 4688 (process creation), 4720/4732 (user/group management)
  - PowerShell logs
- **HTTP & HTTPS**
  - Request and response structure
  - HTTP methods (GET, POST, PUT, DELETE)
  - Status codes
  - TLS/SSL protocols
  - Cookies and session management
- **Protocols & Ports**
  - TCP vs. UDP
  - The three-way handshake process
  - Common ports (22, 80, 443, etc.)
  - Introduction to nmap, Wireshark, and tcpdump
- **Active Directory**
  - Domain, Forest, and Domain Controller concepts
  - Organizational Units (OUs), LDAP, Kerberos, and NTLM
  - Group Policy
  - User and group creation
- **AI in Cybersecurity**
  - Fundamentals of Machine Learning (supervised and unsupervised learning)

- Applications of AI in Security Operations Centers (SOC)
  - Malware detection models
  - Phishing detection techniques
  - Utilization of VirusTotal
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## **Module 0.3: Vulnerability Assessment and Penetration Testing (VAPT)**

### **Learning Objectives:**

This module focuses on identifying, assessing, and exploiting vulnerabilities in systems and applications, with an emphasis on ethical hacking and security testing methodologies.

### **Topics Covered:**

- **Web Application Security**
  - File Inclusion vulnerabilities (LFI, RFI)
  - Cross-Site Scripting (XSS): Reflected, Stored, DOM-based
  - Command Injection
  - SQL Injection
- **OWASP Top 10 (2025)**
  - Broken Access Control
  - Cryptographic Failures
  - Injection attacks
  - Insecure Direct Object Reference (IDOR)
  - Security Misconfiguration
  - Server-Side Request Forgery (SSRF)
- **Nmap Scanning**
  - Live host discovery
  - Basic port scans (TCP, SYN, UDP)
  - Advanced scanning techniques
  - OS fingerprinting
  - Nmap Scripting Engine (NSE) scripts
- **Burp Suite**
  - Proxy setup and configuration
  - Repeater for manual testing
  - Intruder for brute-force attacks (Sniper, Cluster Bomb)
  - Payload configurations
- **Vulnerability Tools**
  - CVE (Common Vulnerabilities and Exposures) and CVSS (Common Vulnerability Scoring System)

- Metasploit framework: Searching and running exploits
- Vulnerability Capstone projects
- **Shell Techniques**
  - Bind, Reverse, and Web shells
  - Netcat and Socat utilities
  - msfvenom payloads
  - Metasploit handler
  - WebShells
- **Privilege Escalation**
  - **Linux:**
    - System enumeration
    - linPEAS tool
    - Kernel exploits
    - Abuse of Sudo, SUID, and GTFOBins
    - Capabilities, Cron jobs, and NFS
  - **Windows:**
    - Password harvesting
    - Service misconfigurations
    - DLL hijacking
    - Token impersonation
    - WinPEAS tool
- **AI / Evil GPT**
  - AI-driven threat actors
  - AI-powered phishing campaigns
  - Automated vulnerability discovery
  - Defense awareness strategies

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## Key Tools — VAPT Module

| Category             | Tools                              |
|----------------------|------------------------------------|
| Exploitation         | Metasploit Framework, Msfvenom     |
| Scanning             | Nmap, Burp Suite, Nikto, OWASP ZAP |
| Privilege Escalation | linPEAS, WinPEAS, GTFOBins         |
| Analysis             | Wireshark, VirusTotal, CyberChef   |

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# Module 0.4: Security Operations Center (SOC)

## Learning Objectives:

This module provides an in-depth understanding of the roles, responsibilities, and technologies used in a Security Operations Center (SOC) to monitor, detect, and respond to security incidents.

## Topics Covered:

- **SOC Fundamentals**

- Roles and responsibilities of L1 and L2 Analysts
- Core functions: monitoring, detection, incident response, and log analysis

- **Security Frameworks**

- Cyber Kill Chain (7 stages)
- MITRE ATT&CK: Tactics, Techniques, Sub-techniques, Navigator, and threat profiling

- **Log Analysis**

- **Windows:**
  - Event IDs: 4624, 4625, 4688, 4720, 4732
  - Security, System, and Application logs
  - Detection of brute force attacks and lateral movement
- **Linux:**
  - Log files: /var/log/auth.log, syslog, secure, messages
  - SSH brute force detection, sudo misuse, cron jobs, and service failures

- **SIEM Tools**

- **Splunk:**
  - Installation and setup of forwarders
  - Architecture: Indexer and Search Head
  - SPL (Search Processing Language) queries, dashboards, alerts, and field extraction
- **ELK Stack:**
  - Elasticsearch, Logstash, and Kibana
  - Advanced ELK queries
  - Kibana dashboards
- **Wazuh:**
  - Agent-based monitoring
  - Vulnerability assessment
  - APL reports

- **Network Monitoring**

- Snort: Sniffer, Packet Logger, IDS/IPS modes
- Rule writing and PCAP analysis

- Wireshark for traffic analysis and incident response
  - **Incident Handling**
    - Phishing analysis using VirusTotal, URLScan, and Any.run
    - Development of playbooks for Phishing, Malware, Brute Force, and Insider Threat scenarios
  - **Threat Analysis**
    - Indicators of Compromise (IOC) analysis
    - Malware behavior analysis
    - Log correlation techniques
    - MITRE mapping
    - Integration of threat intelligence
  - **Cloud Security**
    - Cloud service models: IaaS, PaaS, SaaS
    - Cloud monitoring techniques
    - Identity and Access Management (IAM) security
    - Data protection strategies
    - Cloud Incident Response (IR)
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## SIEM + Monitoring Tools

| Module           | Topic                                                                  |
|------------------|------------------------------------------------------------------------|
| <b>Splunk</b>    | SPL queries, dashboards, threat correlation, alerting                  |
| <b>ELK Stack</b> | Log ingestion, visualization, analysis                                 |
| <b>Wazuh</b>     | Open-source XDR/SIEM, agent-based monitoring, vulnerability assessment |

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## Key Tools and Technologies

### Networking & Security

- Burp Suite, Splunk, Hashcat, Metasploit, Suricata, Aircrack-NG

### VAPT Tools

- **Exploitation:** Metasploit, Msfvenom, SQLmap, Netcat
- **Scanning:** Nmap, Burp Suite, Nikto, OWASP ZAP
- **Privilege Escalation:** linPEAS, WinPEAS, GTF0Bins
- **Analysis:** Wireshark, VirusTotal, CyberChef

## **SIEM & Monitoring**

- Splunk, ELK Stack (Elasticsearch, Logstash, Kibana), Wazuh, Snort

## **AI in Cybersecurity**

- VirusTotal, ML-based threat detection, AI-driven phishing analysis
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